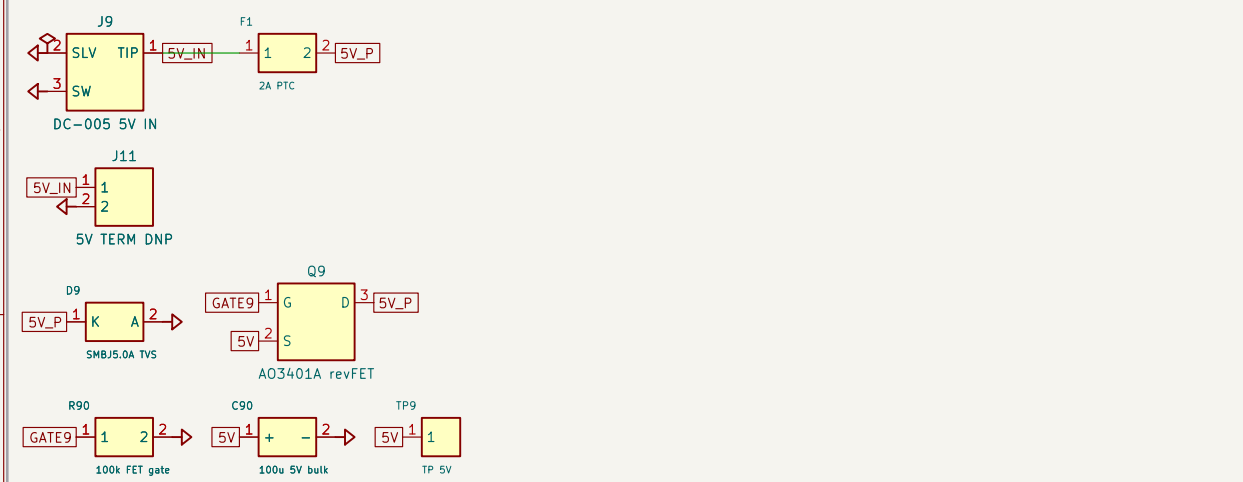
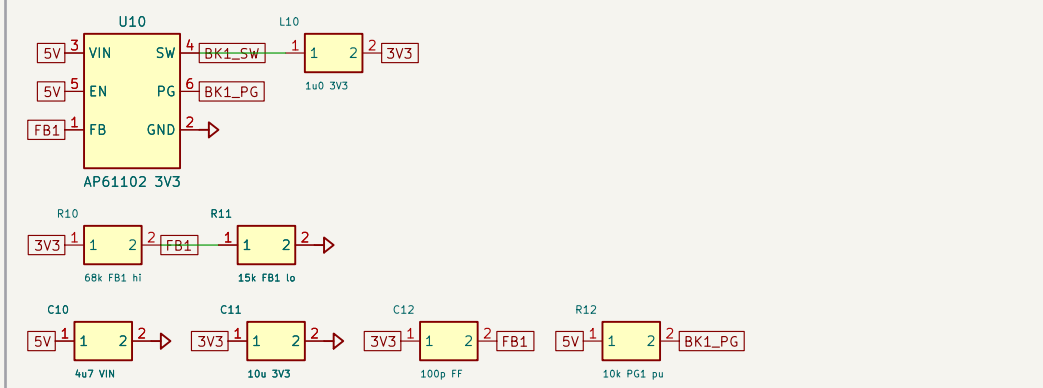


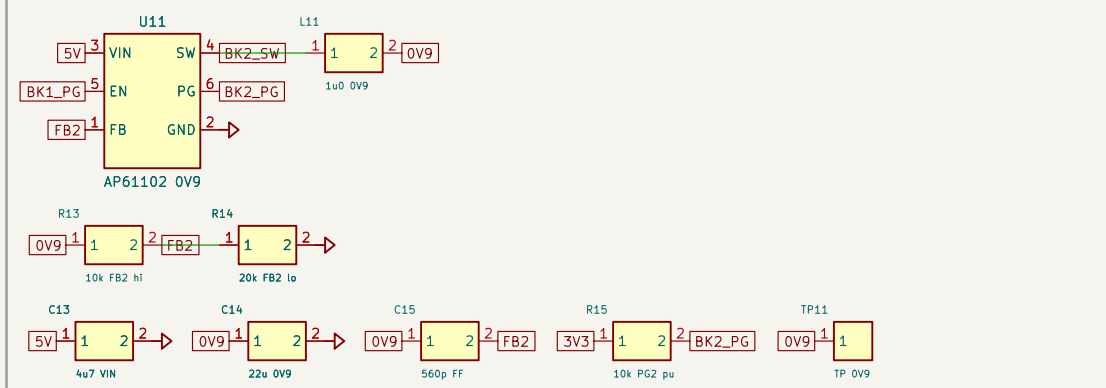
1. 5V ENTRY + PROTECTION: barrel(gop) / terminal(DNP): PTC+TVS+revFET [ADR-0002, 04]



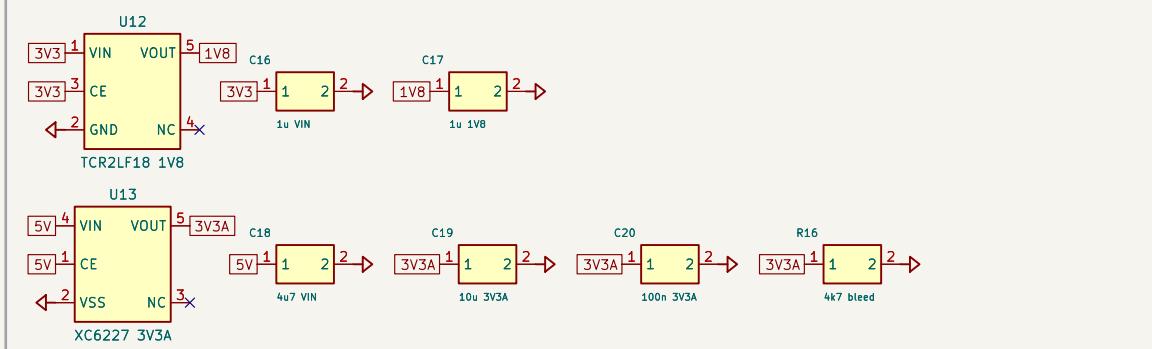
2. BUCK 3V3 (AP61102): VO=0.6(1+68/15)=3.32V; EN=5V always-on [Xmos ref U16]



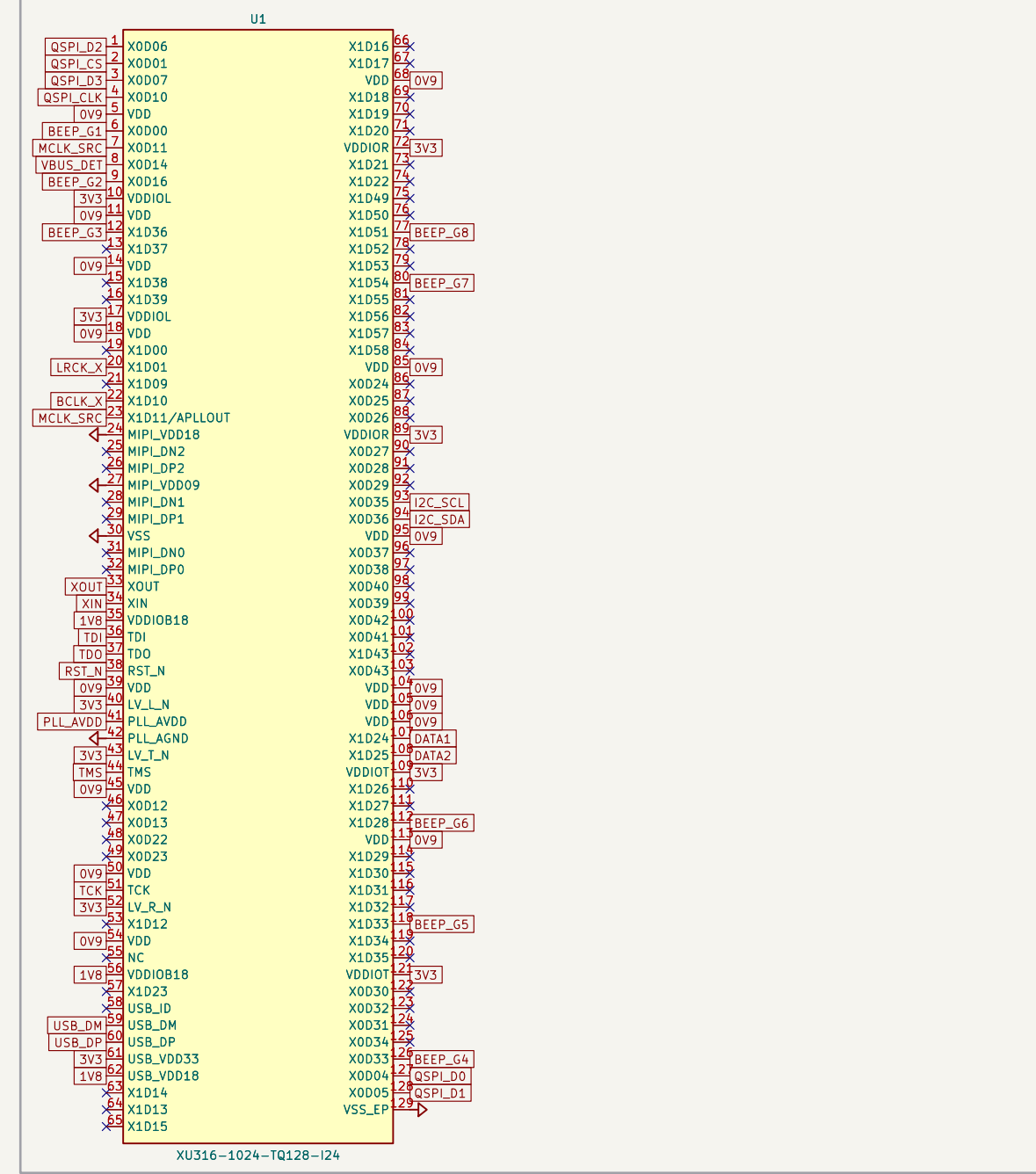
3. BUCK DV9 core (AP61102): VO=0.6(1+10/20)=0.9V; EN<-3V3 P6 (seq) [Xmos ref U17]



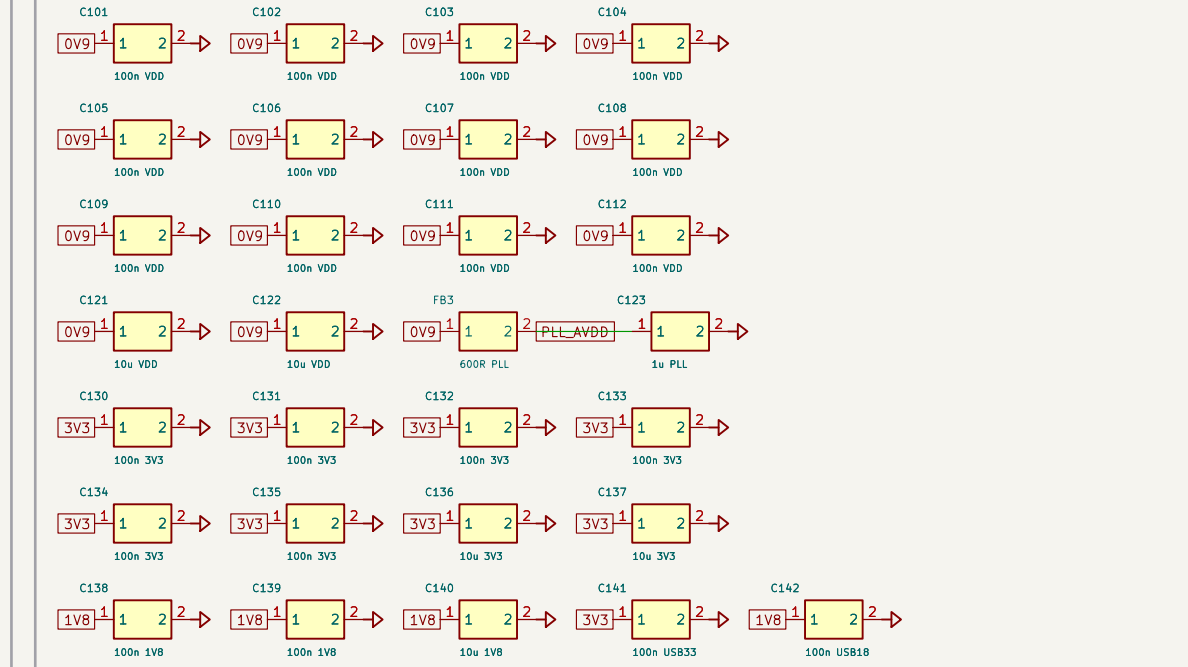
4. LDO 1V8 (TCR2F18 <-3V3) + LDO 3V3A (XC6227 <-5V, quiet analog) [Xmos ref U18/U15]



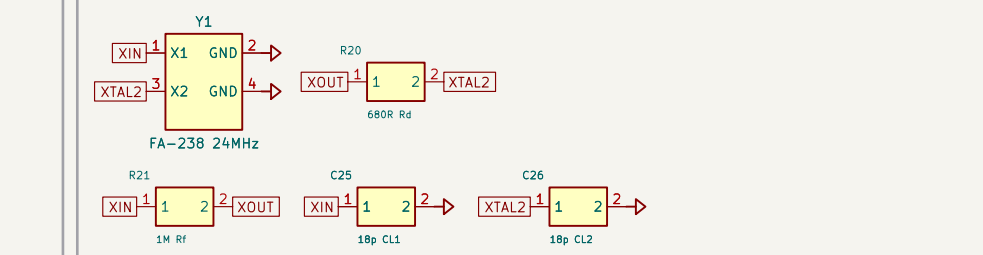
5. XU316 decoupling: 12x100n VDD, per-VDDIO 100n, bulk 10u; PLL_AVDD FB+1u [part.yami H.2]



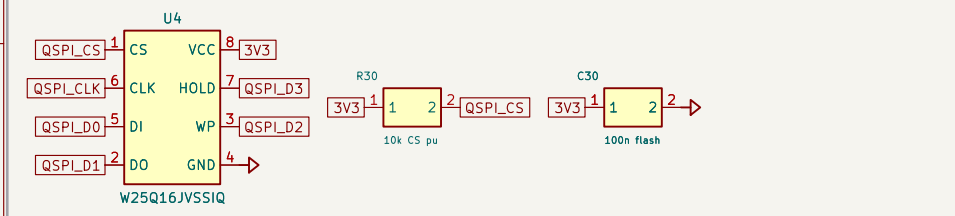
6. XU316 decoupling: 12x100n VDD, per-VDDIO 100n, bulk 10u; PLL_AVDD FB+1u [part.yami H.2]



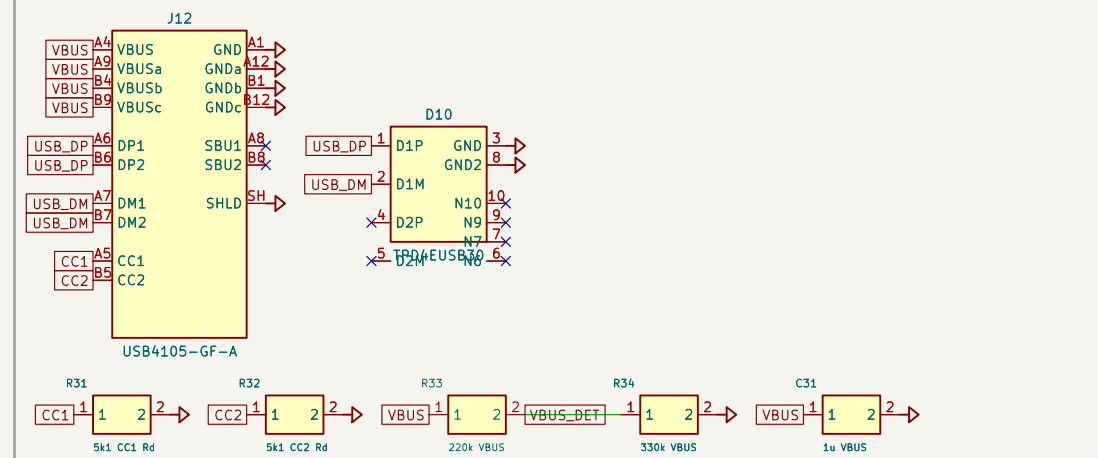
7. 24MHz crystal (FA-230): Rf 1M, Rd 680R, CL 18pF (USB clock, \$7.3 p19)



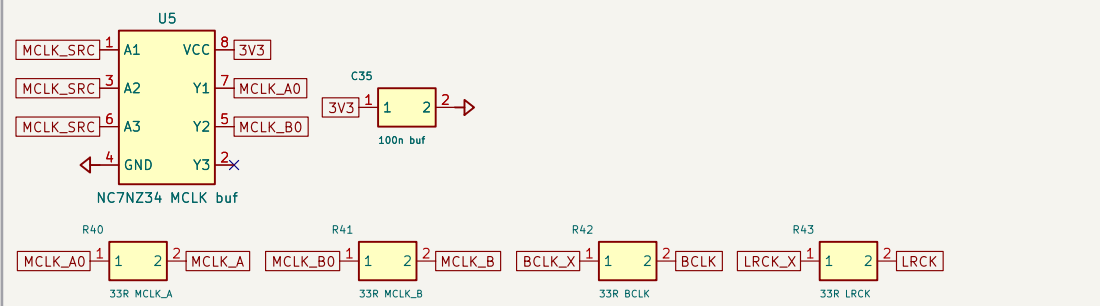
8. QSPI boot flash (W25Q16): CS_N pull-up 10k to 3V3 (Fig 12 p22)



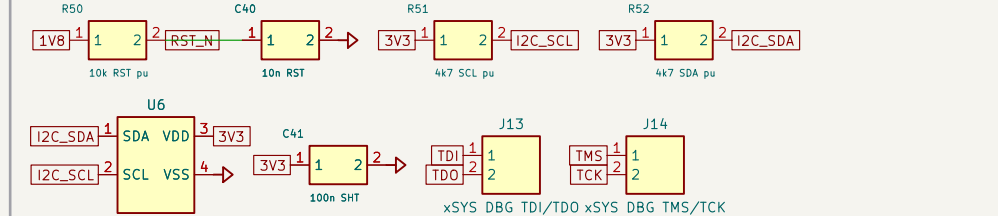
9. USB-C device (USB4105) + ESD (TPD4EUSB30) + CC Rd + VBUS sense [\$11.4/\$14.1]



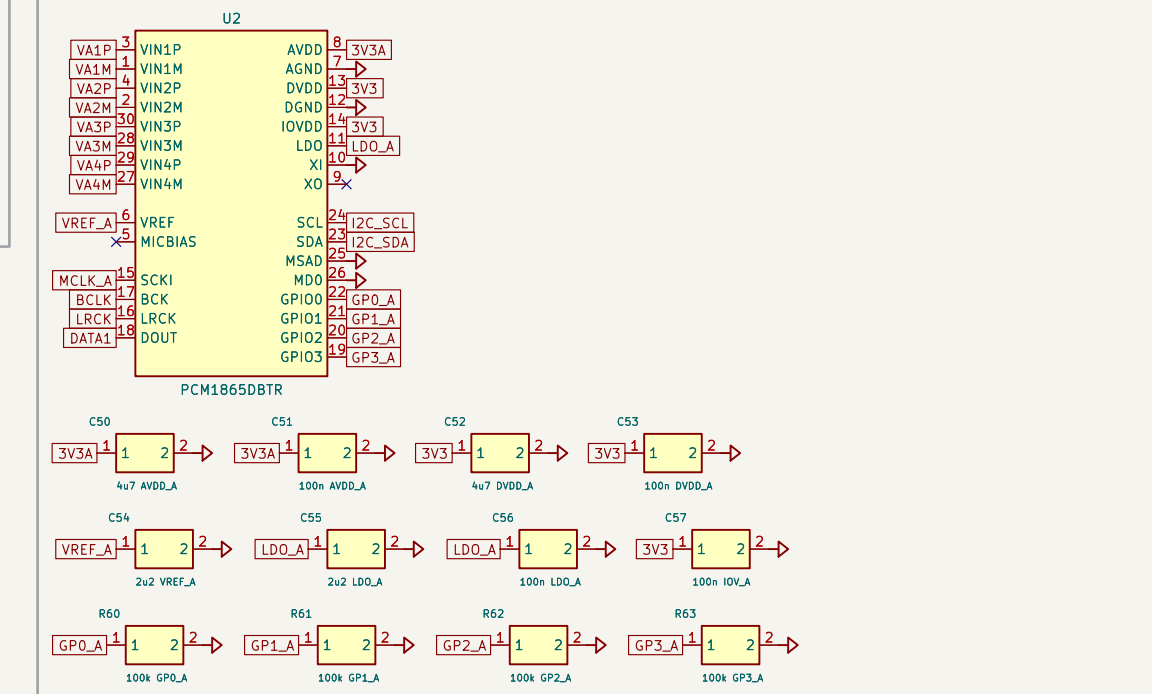
10. MCLK buffer (NC7NZ34) fanout to both ADCs: BCLK/LRCK 33R source term [\$3 clocks]



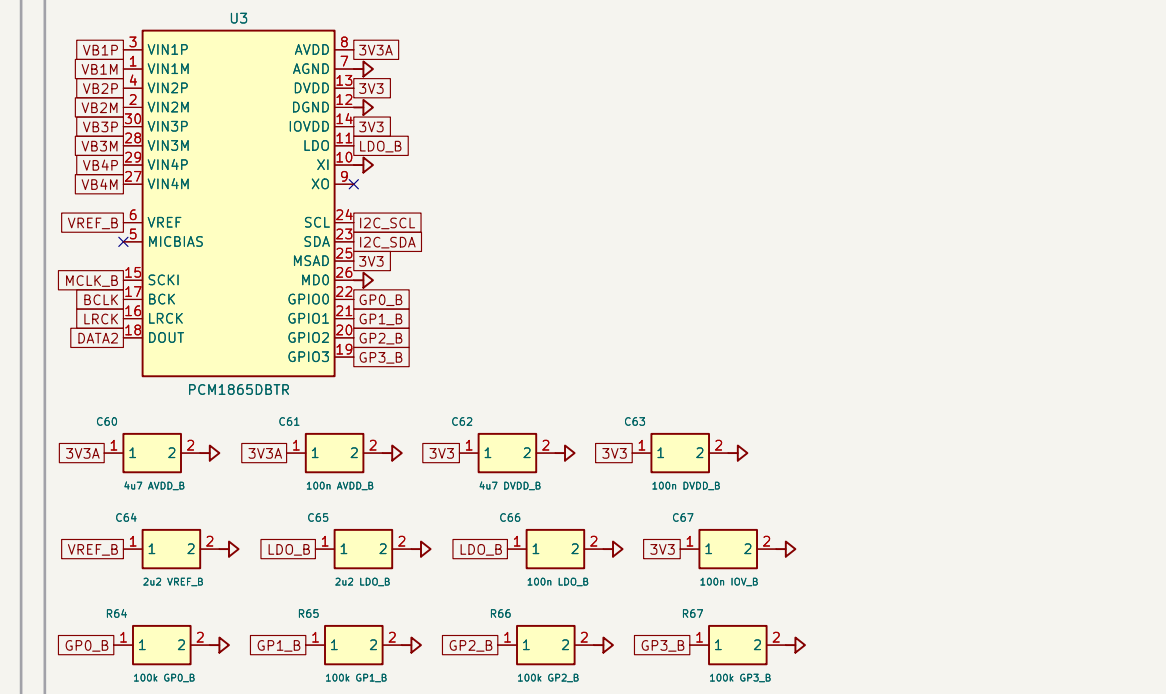
11. RST_N RC (10k->1V8, 10n) + I2C 4k7 pull-ups + SHT40 T/RH [\$2 reset, \$7]



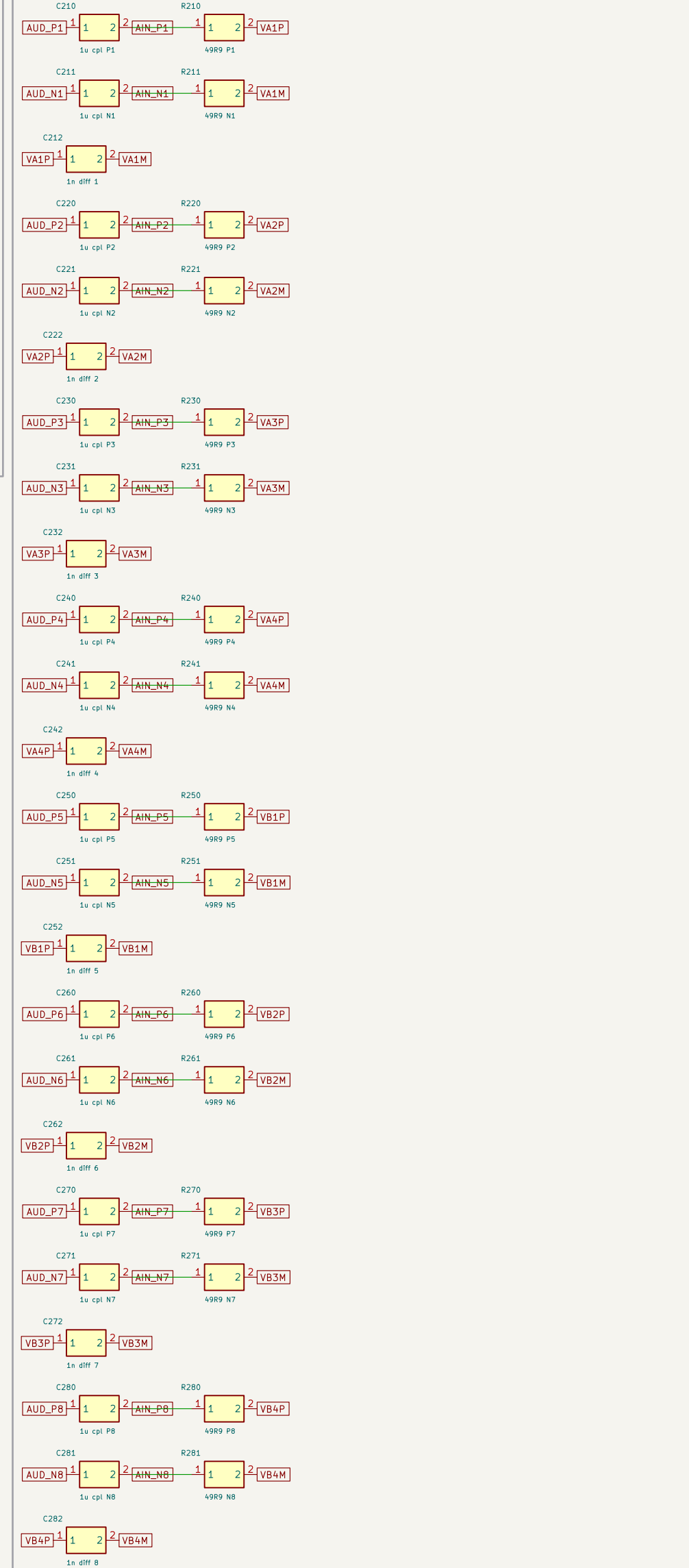
12. ADC1 PCM1865 (ch1-4, I2C 0x4a) + decoupling + GPIO 100k pulldowns [\$5, ADR-0006]



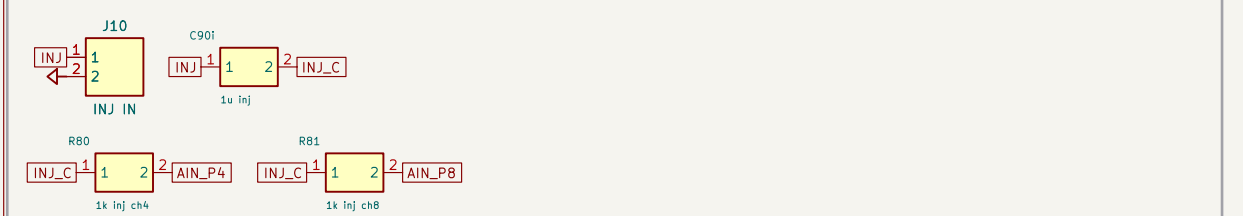
13. ADC2 PCM1865 (ch5-8, I2C 0x4b) + decoupling + GPIO 100k pulldowns [\$5, ADR-0006]



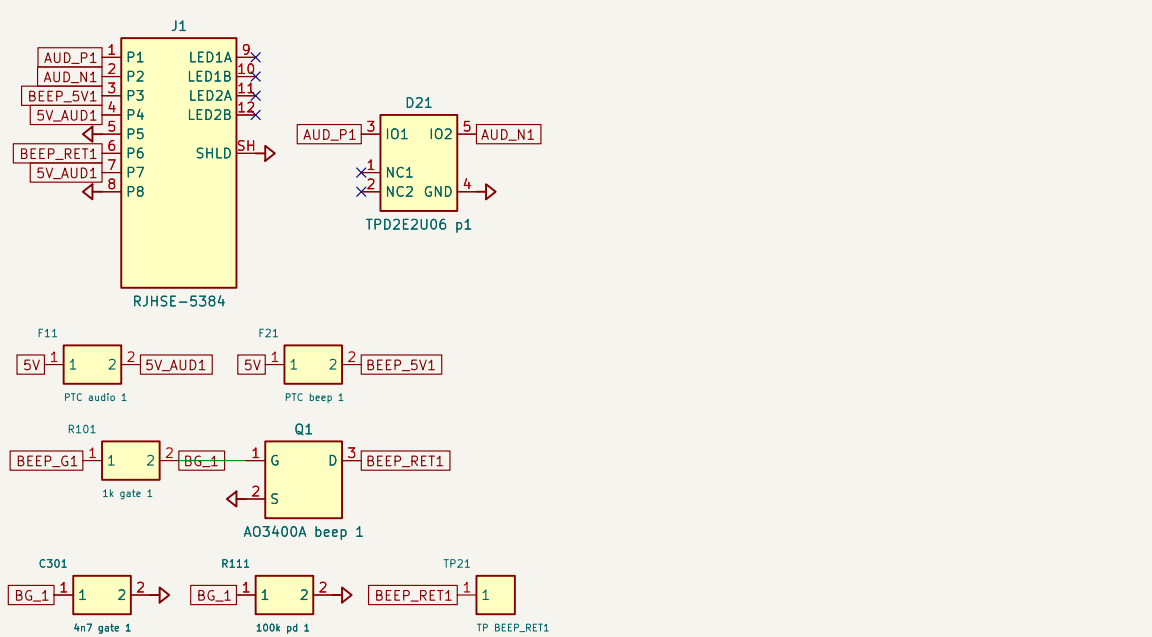
14. ADC INPUT COUPLING (8x): AUD_Pn/Nn -> 1u -> 490R -> PCM1865 diff pair



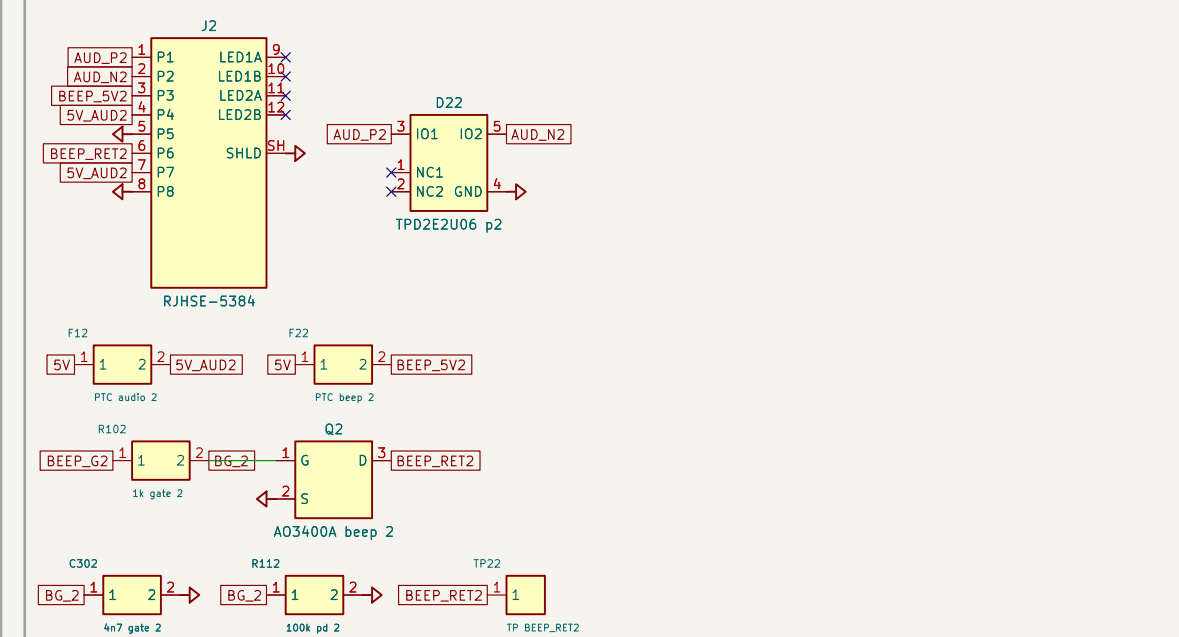
15. SKEW INJECTION HEADER (J10): INJ -> 1u -> INJ_C -> 1k into ch4 & ch8 [D7, \$5A]



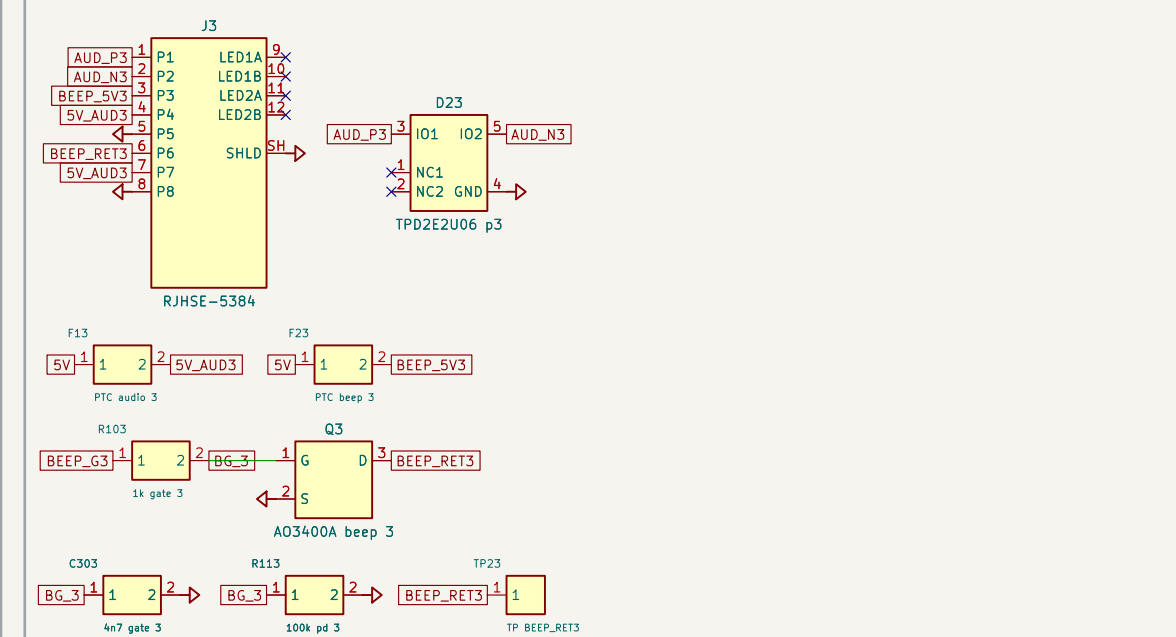
P1. PORT 1 (RJ45 J1) NOT ETHERNET: ESD + audio/beep PTC + slowed beeper FET [ADR-0005]



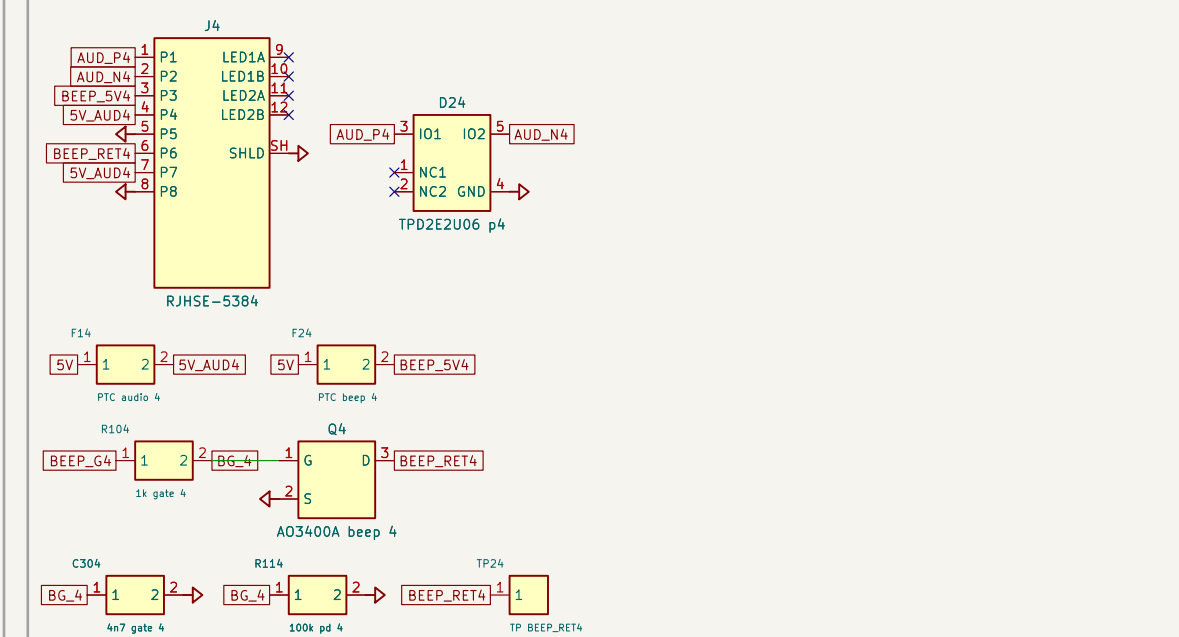
P2. PORT 2 (RJ45 J2) NOT ETHERNET: ESD + audio/beep PTC + slowed beeper FET [ADR-0005]



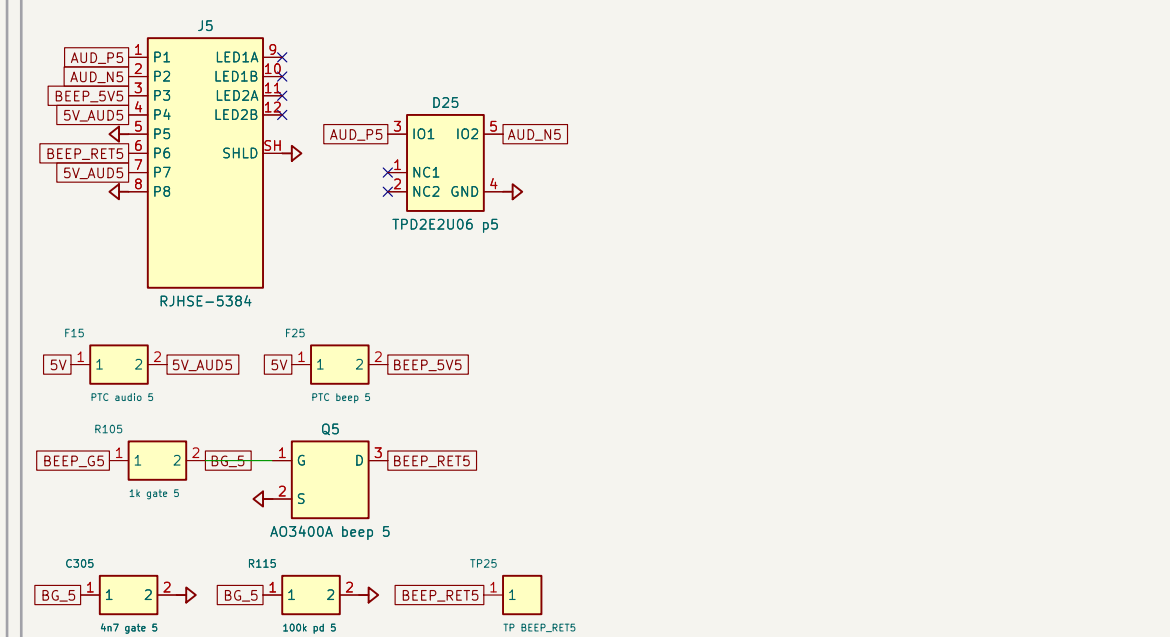
P3. PORT 3 (RJ45 J3) NOT ETHERNET: ESD + audio/beep PTC + slowed beeper FET [ADR-0005]



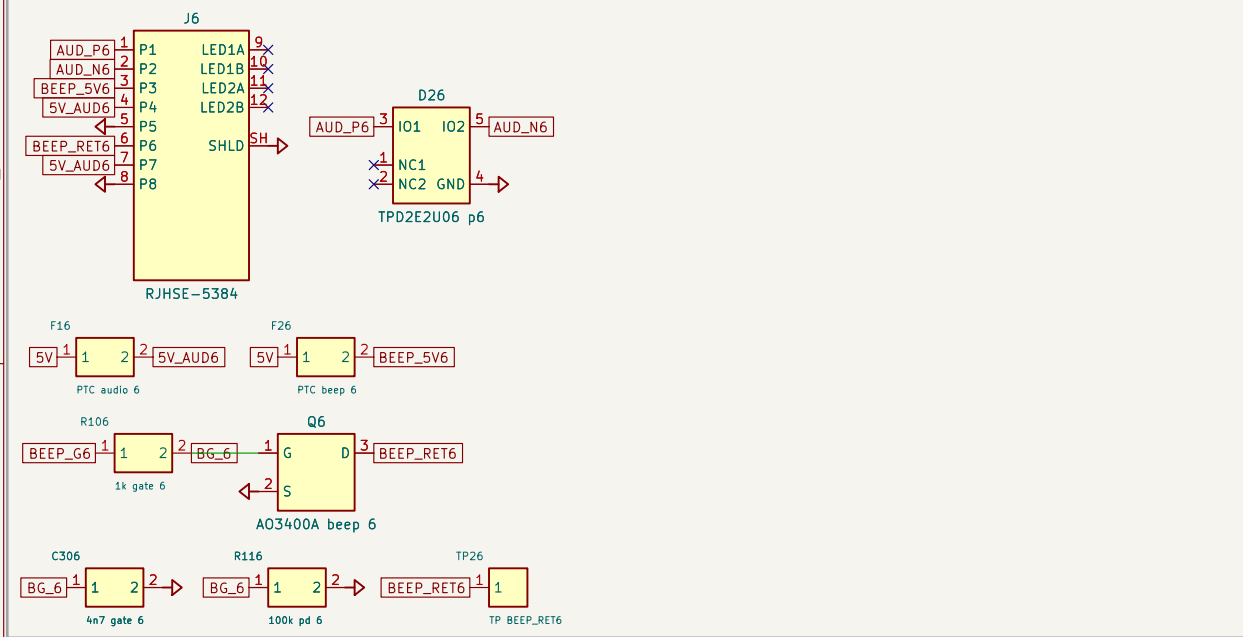
P4. PORT 4 (RJ45 J4) NOT ETHERNET: ESD + audio/beep PTC + slowed beeper FET [ADR-0005]



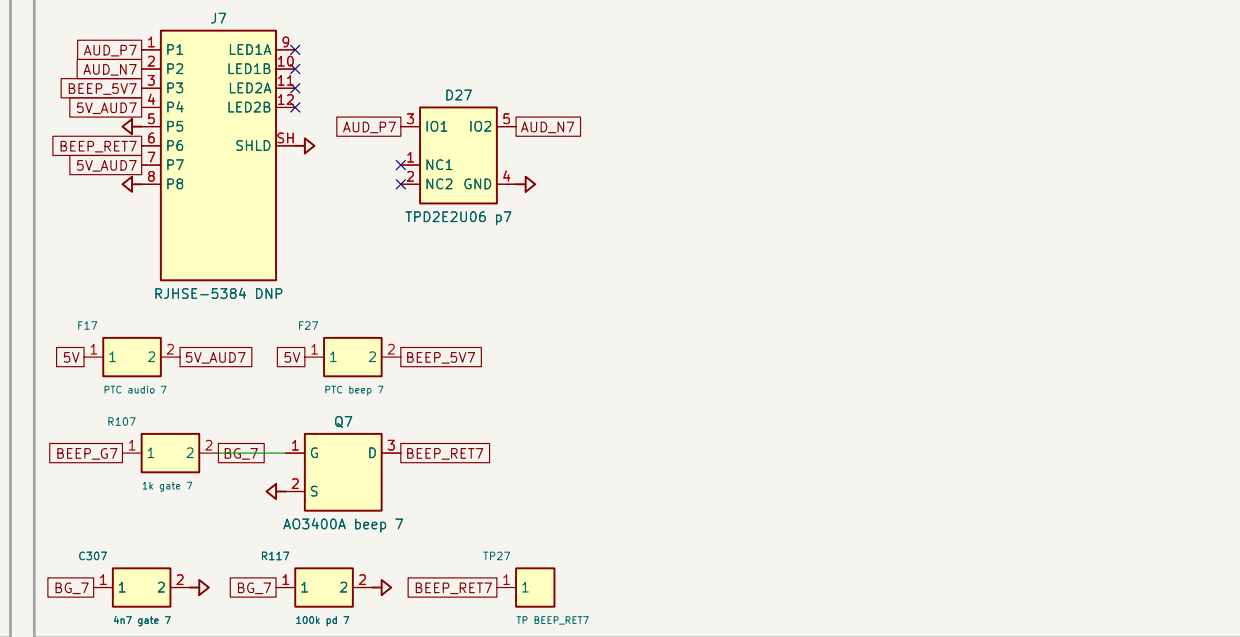
P5. PORT 5 (RJ45 J5) NOT ETHERNET: ESD + audio/beep PTC + slowed beeper FET [ADR-0005]



P6. PORT 6 (RJ45 J6) NOT ETHERNET: ESD + audio/beep PTC + slowed beeper FET [ADR-0005]



P7. PORT 7 (RJ45 J7 DNP) NOT ETHERNET: ESD + audio/beep PTC + slowed beeper FET [ADR-0005]



P8. PORT 8 (RJ45 J8 DNP) NOT ETHERNET: ESD + audio/beep PTC + slowed beeper FET [ADR-0005]

